

Johnson C. Smith University
Topics in Algebra | MTH 439 A | Spring 2014
Tuesday & Thursday 12:00-1:15 | MAIN SHB 303

"Mathematics is the art of giving the same name to different things."

~Henri Poincare, responding to "Poetry is the art of giving different names to the same thing."

In our own time, in the period 1960-1980, we have seen particle physics emerge as the playground of group theory.

~Freeman Dyson

Instructor Information

Name: Dr. James Ashe

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Office hours: TR 8:30-9:45 and 10:45-12; TR 4:15-7:45 (by appointment)

Course Description

This is a **3 credit hour** course. Topics on semi-groups and groups, rings, integral domains, finite fields, modules over ring, and vector spaces over fields. Prerequisite: MTH 335. \

Course Objectives

The main objective of this course is to develop mathematical reasoning by learning how to read, write, and present proofs. Students completing this course will demonstrate knowledge of the fundamental concepts in the core areas listed in the above section. Additionally, students should demonstrate the ability for abstract reasoning, rigor, and logical thinking.

Instructional methods will consist of lectures, in-class discussions, exams, and student presentations. Other methods may be adopted according to the particular needs of the class.

Course Materials

Suggested Textbook: There is no required textbook. Below is a list of recommended reference books. The first three are free.

Abstract Algebra: Theory and Applications *by Thomas W. Judson*

A Gentle Introduction to Abstract Algebra *B. A Sethuraman*

Abstract Algebra: The Basic Graduate Year *by Robert Ashe*

Contemporary Abstract Algebra *by Joseph A. Gallian*

Grades

Grading scale: $A \geq 90 > B \geq 80 > C \geq 70 > D \geq 60 > F$

Evaluation: Grades will be taken from homework, quizzes, and Exams as follows:

Homework= 40%; Presentations= 20%; Midterm= 20%; Final = 20%

Homework: Regular homework assignments will be given. Students can collaborate on homework assignments, however each student should hand in their own assignments. The lowest homework score will be dropped.

Presentations: Each student will be given one problem to present to the class from each homework assignment. The date of presentation will typically precede the homework due date. You will be assessed on both your ability to work the problem and explain it.

Midterm and Final Exam: Both the midterm and final will be comprehensive exams; see the schedule below. *If beneficial, the final exam score will replace the midterm exam score.*

Class Policies

JCSU policies regarding attendance, academic integrity, grades, and change of enrollment will be enforced. It is the student's responsibility to be aware of these policies and of related deadlines as well as any announcements or syllabus adjustments (written or oral) made in class. Please see the student handbook for details, jcsu.edu/docs/handbook.pdf.

Attendance: Johnson C. Smith University has no official attendance policy. However, because attendance in classes is a vital part of the educational process, students are encouraged to attend classes regularly and promptly.

Students who may miss classes while representing the University in an official capacity are exempt from regulations governing absences. However, absence from class for official University business does not relieve the student from responsibility for any class assignments that may be missed during the period of absence. In such cases, *it is the student's responsibility to make arrangements to make up any work that will be missed before the assignment/quiz/exam is missed.*

Late Homework: An assignment will have 25% deducted for each day it is late.

Missed presentations: A student will be allowed one make-up for a missed presentation. Presentations can be rescheduled if a University sponsored event conflicts with the presentation date, and advance notice of the absence was given by the student.

Missed exams: Missed exams may be made up at the discretion of the Instructor. (1) you must notify me by email or phone within 24 hours of the missed exam AND (2) you must have no more than 3 unexcused absences in the class. If BOTH of these conditions are met, a make-up will be given. However unless a documented excuse is provided, 10% will automatically be deducted from the make-up. If you will be gone for any reason, it will be *your* responsibility to inform me at least a week in advance to schedule to take the exam early.

Other

Important Dates

Friday 17 th January, 2014	Last day to add/drop
Tuesday 4 th March, 2014	Midterm
Wednesday 1 st May, 2013	Final Exam (10-12pm)

JCSU's academic calendar can be found here: jcsu.edu/happenings/academic-calendar, and the exam schedule can be found here: <http://www.jcsu.edu/happenings/exam-schedule/spring-exam>

Student support services: If you need course adaptations or accommodations for a documented disability please contact the Office of Disability Services.

jcsu.edu/directory/federal-requirements/general-institutional-information/jcsu-disability-services,
phone: (704) 398-1116.

The Office of Counseling Services provides confidential intakes and assessments, personal counseling and referrals to appropriate resources to all enrolled students on a walk-in or appointment basis.

jcsu.edu/admissions/future_students/meet_your_counselor, phone (704) 378-1044.

Disclaimer: While I do not foresee any significant changes to the items in this syllabus during the semester, I reserve the right to make appropriate changes. Students will be notified of any such changes in writing.

You know, for a mathematician, he did not have enough imagination. But he has become a poet and now he is fine.

~David Hilbert

If you want to put the holes in your knowledge on display, try teaching someone.

~Linux guru Alan Cox